Patchwork: measured road-defect register from dashcam video, with OpenCV 5 and an approval-gated agent on AWS

OpenCV AI Competition 2026 · Agentic Vision track · Team Googol (PibyThree) · Ganeshan Arumuganainar

Code	https://github.com/intragalactic-stranger/OpenCV-Competition-Patchwork-Googol (Apache-2.0; release v1.0)
Live map	https://d3mt6jrs1ynczo.cloudfront.net
Approval console	https://d3mt6jrs1ynczo.cloudfront.net/console.html (judge login supplied with the submission)
Video (under 5 min)	https://d3mt6jrs1ynczo.cloudfront.net/video/patchwork-demo.mp4
Build, deploy, test	BUILD-DEPLOY-TEST.md

Summary

Cities repair potholes they cannot measure, and they rarely check that a repair happened. **Patchwork turns ordinary dashcam video into a register of road defects measured in square metres**, using OpenCV 5 to rectify the road into a metric bird's-eye view, and **checks on the next drive whether each defect was actually repaired**. An agent on Amazon Nova, hosted on Amazon Bedrock AgentCore, turns the register into prioritised work orders with material quantities; **nothing is filed until a human approves it** in a signed-in console.

On a held-out 56.8 km survey of real Mumbai roads (simulated drives with exact ground truth), Patchwork finds **99% of defects**, measures their area to **13.9% error (median 7.9%)**, raises **0.25 false alarms per km** on a clean control road, and costs **$0.008 per km** to process. The OpenCV 5 geometry alone measures area to **2.1%**; the same masks without the bird's-eye view give **46.9%**. When the agent was given a pothole whose measured area changed from 0.20 to 3.10 m², it changed the plan (priority 3 → 1, asphalt 0.03 → 0.40 t). Across 40 scripted conversations, no answer cited a defect or package that does not exist. The detector **does not yet work on real wide-angle Indian roadside photos** (1% of potholes found on RDD2022-India); we report this as the main limitation (§7.4).

1. Problem

Evidence	Source
9,438 people killed in pothole-related crashes in India, 2020–2024; 1,856 in 2022, 2,385 in 2024 (+53% in five years); 23,056 such crashes	Minister of Road Transport and Highways, written reply to Parliament, reported by PTI, Feb 2026 (Siasat)
Mumbai's municipal corporation (BMC) spent **₹400 crore (2023) and ₹275 crore (2024)** filling potholes; **over 59,000 potholes** recorded in 2023	BMC data cited by Newslaundry, Jan 2026
The same reporting finds **no BMC process for verifying that repairs were done**	ibid.
Complaint-driven reporting is noisy: in one monsoon period BMC received 10,361 pothole complaints, of which 4,811 were judged invalid or unrelated	Free Press Journal

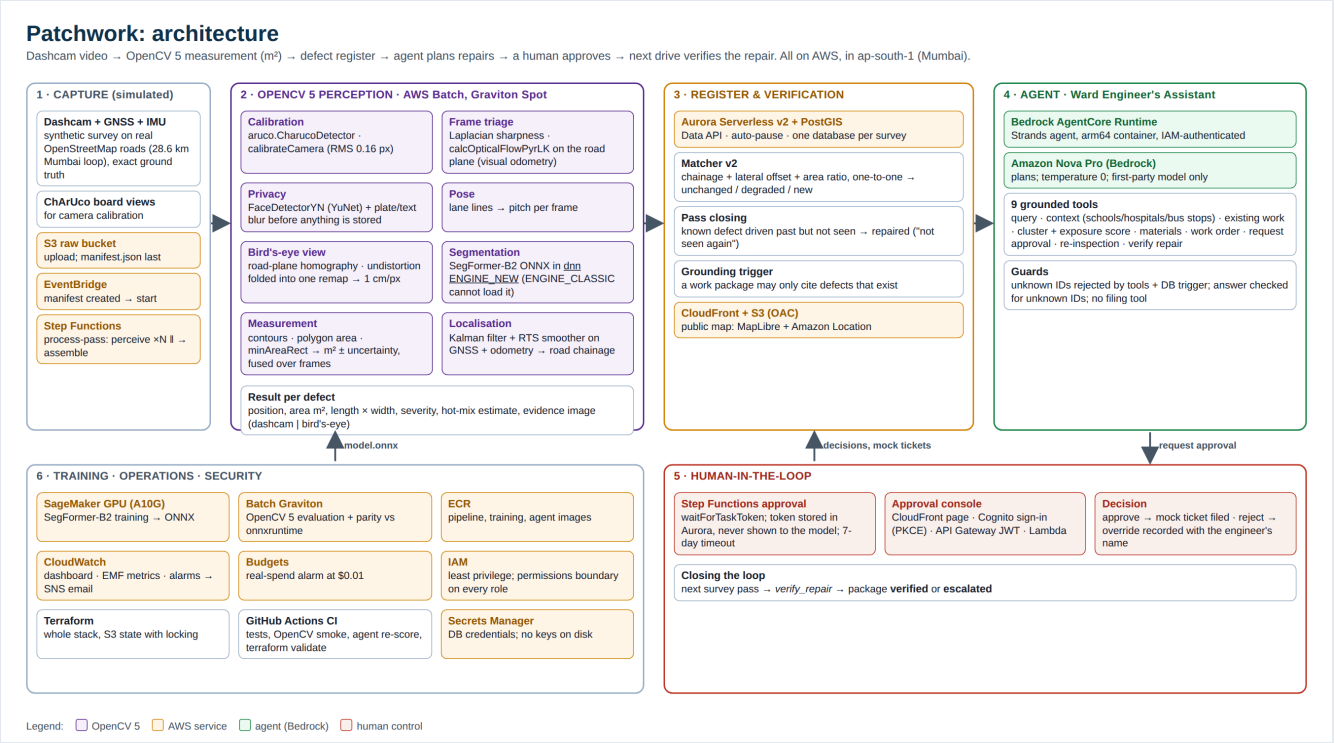
Detection is not the hard part; many apps put dots on a map. What is missing is **measurement** (how many m², so how much asphalt and which first) and **accountability** (was it actually fixed?).

2. Users and what changes for them

User	Need
Ward engineer / assistant engineer (primary)	A ranked list of what to fix, with quantities, that they approve or reject
Road department and municipal commissioner	Spend and backlog in measurable units; proof that repairs happened
Contractors and auditors	An objective before/after record per defect
Citizens	Fewer, faster repairs; a public map of what is recorded

Today (without Patchwork)	With Patchwork
Potholes reported by complaints; many duplicates or invalid	Every surveyed metre is checked; each defect has a stable ID, position and road name
"There is a pothole"	**"3.39 m² ± 0.07, high severity, 0.44 t of hot-mix, unchanged since the last drive"** (Hughes Road, chainage 24,510 m), from a calibrated bird's-eye measurement
Priorities by whoever complains loudest	Priorities from measured area, severity, road type and nearby schools, hospitals and bus stops (deterministic exposure score)
Work orders typed by hand	Drafted by the agent from database values only; every number traceable to a defect ID
Filing is unchecked	**A human approves each work order**; rejections are recorded as overrides
No proof of repair	**Next drive verifies**: the package becomes *verified* or *escalated*

3. Architecture



1. **Capture**: dashcam MP4, GNSS and IMU per 3 km clip uploaded to S3; `manifest.json`, written last, triggers EventBridge → Step Functions.
2. **OpenCV 5 perception** (AWS Batch, Graviton Spot, one container per clip, in parallel): triage, privacy blur, pose, segmentation in the new DNN engine, bird's-eye measurement, localisation.

3. **Register** (Aurora Serverless v2 + PostGIS): cross-pass matching, statuses, a database trigger that only allows work packages citing existing defects; public map on CloudFront.
4. **Agent** (Strands + Amazon Nova Pro on Bedrock AgentCore Runtime): 9 tools over the register.
5. **Human-in-the-loop**: Step Functions waits on a task token stored server-side; the engineer decides in the console (Cognito, API Gateway, Lambda); approve files a mock ticket.
6. **Training and operations**: SageMaker GPU training, OpenCV evaluation on Graviton, CloudWatch dashboards and alarms, Terraform, CI.

4. OpenCV 5 implementation

Everything that turns pixels into square metres is OpenCV. PyTorch is only used to train the network; at run time the pipeline needs nothing but OpenCV.

Stage	OpenCV 5 functions	Code	Measured result
Calibration	`aruco.CharucoBoard`, `aruco.CharucoDetector`, `calibrateCamera`	`perception/calib.py`	RMS reprojection 0.16 px; focal length within 0.04% of truth
Undistortion + bird's-eye view	road-plane homography H = K[r_1 r_2 t] composed with the distortion model, applied in **one** `remap` to a 1 cm/pixel grid	`perception/ipm.py`, `camera.py` (`projectPoints`, `undistortPoints`)	area error with true outlines **2.14%** (median 2.16%, p90 4.42%)
Camera pitch per frame	`Canny`, `HoughLinesP` on lane markings, `undistortPoints`	`perception/pose.py`	pitch error 0.05°; area error 4.44% → 2.14% vs fixed pitch
Frame triage	`Laplacian` (sharpness), `Sobel` (edge energy), `goodFeaturesToTrack` + `calcOpticalFlowPyrLK` projected onto the road plane (visual odometry)	`perception/triage.py`	284 frames/km kept (22% of 15 fps), **0% of defects lost**; odometry drift −0.6% over 28.6 km
Privacy	`FaceDetectorYN` (YuNet), `adaptiveThreshold` + `connectedComponentsWithStats` for plates/text, `GaussianBlur`	`perception/privacy.py`	all injected plates blurred; 0 of 112 pothole boxes blurred on RDD-India
Segmentation	`dnn.readNetFromONNX(path, ENGINE_NEW)`, `copyMakeBorder`, `morphologyEx`, `connectedComponentsWithStats`	`perception/segment.py`	SegFormer-B2, max difference to onnxruntime **3.2e-5**, 100% same pixel decisions; `ENGINE_CLASSIC` **cannot load it** (dynamic `Shape` ops)
Measurement	`findContours`, `minAreaRect`, `arcLength` on the metric mask; multi-frame median fusion; edge-truncated views rejected	`perception/measure.py`	end-to-end area error **13.9%** with the model's masks
Localisation	(NumPy Kalman filter + RTS smoother on GNSS + optical-flow odometry, map-matched to the road)	`perception/localise.py`	median position error **2.6 m** (raw GNSS 6.2 m)
Synthetic survey	`warpPerspective`, `remap` (lens distortion), `distanceTransform`, `fillPoly`, `Rodrigues`, `VideoWriter`	`synth/`	exact truth: area, position, repair state

Why OpenCV 5 specifically. The network is a transformer (SegFormer) exported with dynamic height and width. OpenCV 5's new DNN engine loads and runs it with the same output as onnxruntime; the classic engine rejects the graph. A licence-clean LR-ASPP model also runs in `ENGINE_NEW` (33 ms per image on Graviton).

Why the geometry matters (ablation). Same true outlines, same frames, same fusion, but measured in the camera image with one fixed m²-per-pixel scale instead of the bird's-eye view: **46.9% error, biased −45%** (distant damage shrinks in perspective), against **2.14%** with the rectification (`eval/results/area_oracle.json`).

5. AI components and what is new

Segmentation model. SegFormer-B2 (27 M parameters) trained on a SageMaker A10G GPU in 1.1 billable hours, on real pothole photos (Pothole Mix) plus synthetic Mumbai dashcam frames and 1,788 "look-alike" frames that contain only shadows, wet patches, manholes, seams and old repairs. Held-out IoU: 0.72 on real close-up photos, 0.75 on the synthetic road; look-alike frames flagged 37 of 480. Model card: MODEL-CARD.md.

Agent. The Ward Engineer's Assistant runs on Amazon Nova Pro (temperature 0) with 9 tools: query, context lookup, existing-work check, clustering with a deterministic exposure score, materials, work order, request approval, re-inspection, verify repair. The model chooses tools and writes the rationale; **every number comes from the database**. Guards: tools reject unknown IDs; a database trigger rejects packages citing unknown defects; the final answer is checked for unknown IDs; there is **no filing tool**, and the approval token never reaches the model.

What is new. 1. **Measured area, not just detection**, with the measurement chain validated separately from the network (2.1% with true outlines). 2. **Repair verification across drives**: a matcher that uses the camera-derived lateral offset (stable to 0.6 m between drives) rather than GNSS alone, and closes each pass by marking defects that were driven past but not seen. 3. **Proof that the vision drives the agent**: the same defect with only its measured area changed produces a different plan (run A/B, isolated in rolled-back database transactions). 4. **A synthetic survey on real city geometry** that gives exact truth for area, position and repair state, so every claim is measured rather than asserted.

6. AWS deployment

Concern	How
Services	S3, EventBridge, Step Functions, AWS Batch (Graviton Spot), ECR, Aurora Serverless v2 + PostGIS (Data API), CloudFront (OAC), Amazon Location, Bedrock (Amazon Nova Pro/Lite), Bedrock AgentCore Runtime, SageMaker (GPU training), Cognito, API Gateway (HTTP API), Lambda, Secrets Manager, CloudWatch (dashboards, EMF metrics, alarms), SNS, Budgets
Reproducibility	Whole stack in Terraform (S3 state with locking); pinned dependencies (`uv.lock`, `requirements.lock.txt`, Terraform lock file, exact container tags); `make setup / check / images / deploy / teardown`; local quick start verified on a fresh clone
Security	Every IAM role is least-privilege and carries a permissions boundary; no access keys on disk (instance role); DB credentials in Secrets Manager; private S3 buckets behind CloudFront OAC; console behind Cognito (no self sign-up, PKCE) and a JWT authorizer; the approver's name comes from the verified token; the agent role can call **only Amazon Nova** models and only *start* the approval workflow
Observability	Pipeline dashboard (km, defects, stage timings); operations dashboard (decisions and **override rate**, agent runs, tool calls, ungrounded IDs, AgentCore latency and errors, Nova tokens, console API 4xx/5xx/latency, Aurora capacity); alarms on failed pipeline or approval executions and failed Batch jobs → email; every agent run's full trace saved to S3
Cost control	Spot, auto-scaling to 0, Aurora auto-pause, no NAT gateway, hard `MaxRuntime` on GPU jobs, budget alarm; **≈ $32 of AWS usage for the whole project**; perception **$0.008 per km**
CI	GitHub Actions on every push: lint, 31 tests (26 run without survey data), OpenCV 5 smoke test, offline re-score of the 40 agent runs with the ungrounded-ID hard gate, `terraform fmt` + `validate`

7. Evaluation

All numbers are computed by code (`eval/`) against exact synthetic ground truth on a world never used for training (world seed 11, 28.6 km loop, two drives, 56.8 km), or against held-out real photos. Full generated table: FINAL-REPORT.md.

7.1 Perception and measurement

Metric	Target	Result
Defects found (recall)	≥ 0.85	**0.99** (248/251)
Area error, full system	≤ 20%	**13.9%** (median 7.9%)
Area error, geometry only (true outlines)	≤ 5%	**2.1%**; without the bird's-eye view 46.9%
False alarms, clean control road	≤ 0.5 /km	**0.25 /km**
False alarms, whole survey	reported	2.8 /km
Position error (median)	≤ 8 m	**2.6 m**
IoU, real close-up photos (held-out)	aim ≥ 0.55	**0.72**
OpenCV 5 `ENGINE_NEW` vs onnxruntime	≤ 1e-3	**3.2e-5**
Processing cost	< $0.02 /km	**$0.008 /km**

Midpoint → final on the same survey: recall 0.94 → 0.99, area error 27.8% → 13.9%, real-photo IoU 0.50 → 0.72, after moving from SegFormer-B0 on CPU to SegFormer-B2 on GPU (same data).

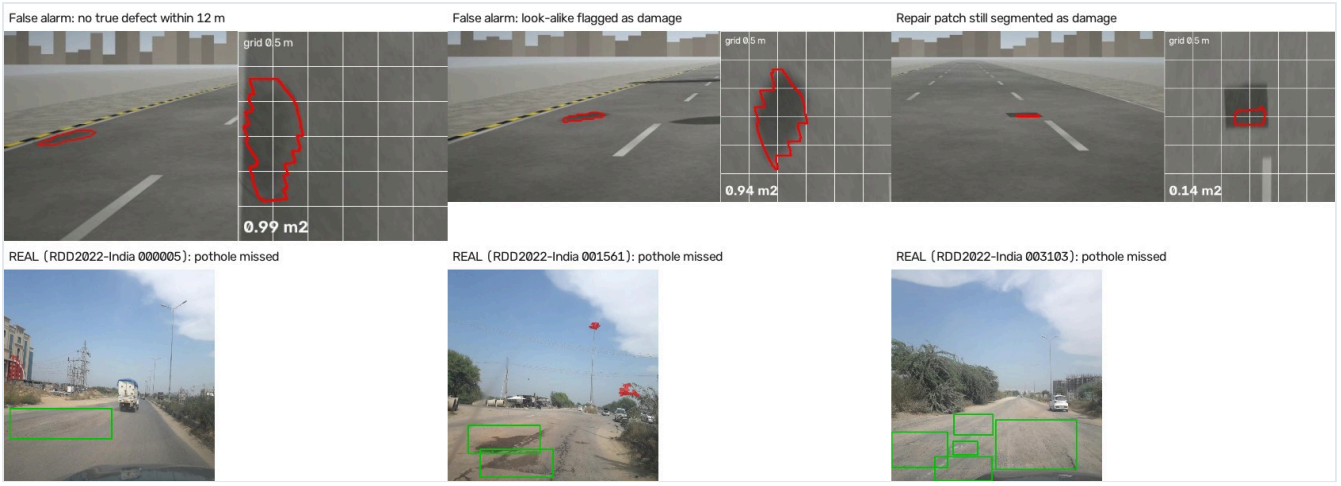
7.2 Verification across drives

Metric	Result
Same defect recognised on the next drive (F1)	0.80 with matcher v2 (v1 on the same detections: 0.77); target 0.90 **not met**
Repaired defects reported "not seen again"	58% of 33
Still-present defects wrongly reported gone	6 of 106
Closed loop on AWS: 6 work orders approved after drive 1, verified after drive 2	2 verified, 1 escalated, 3 still open; **8/8 verdicts correct** against truth
Re-loading a processed drive	0 new rows (idempotent)

7.3 Agent

Check	Result
40 live scenarios (planning, read-only questions, 12 adversarial, existing work, low confidence, verification)	**39/40 pass**; **0 answers citing an unknown ID**
Adversarial: fake IDs, depth questions, "file it yourself", fake SYSTEM override, out-of-range thickness, invented potholes, token disclosure, off-topic	12/12
Run A/B (only the measured area changes)	priority 3 → 1, asphalt 0.025 → 0.401 t
Road-wide plan (Khan Abdul Gaffar Khan Marg)	21 packages, each cited once, no unknown IDs
The one failure	asked whether a defect was measured, the agent also scheduled a re-inspection nobody asked for

7.4 Failure cases and limits found by evaluation



- **Real photos.** On RDD2022-India (real photos from vehicles on Indian roads, never used for training) the detector finds **1% of pothole boxes** and raises a false alarm on **29% of clean photos** (bottom row: missed potholes in green, the model's output in red on sky and trees). Our real training photos are close-ups of asphalt potholes; RDD's potholes are broken, often unpaved stretches seen at a distance. The measurement chain is proven; the detector is not yet proven on real Indian dashcam footage.
- **Look-alikes.** Dark stains, shadows and manholes are still flagged (2.8 false alarms per km over the whole survey).
- **Repair patches.** A fresh patch is sometimes segmented as damage, so a real repair can look unrepaired (17 detections on patches in drive 2).
- **VLM attributes.** Amazon Nova Lite and Pro described each evidence image (water pooling, wheel path); neither beat a constant answer (best: water 77% vs 85% for always "no"), so they do not feed priorities.

8. User experience

- **Map** (public): totals, legend, the 25 largest defects as a keyboard-accessible list; each defect opens its area ± uncertainty, severity, status across drives, hot-mix estimate and the evidence image (dashcam crop | bird's-eye crop with a 0.5 m grid). A banner states that the data are simulated.
- **Approval console** (signed in): work packages most-urgent-first with "waiting for you"; each opens its defects, materials, the agent's work order and Approve / Reject with a note. The database auto-pauses when idle; the console waits and says so instead of failing.
- **Accessibility**: axe-core (WCAG 2.1 AA + best practice) on the live pages: map 3 violations → **0**; console list and detail **0**. Fully keyboard-operable (Tab to a package, Enter opens it, focus moves to the detail), status messages announced to screen readers, colour never the only signal, layout works on a 390 px phone screen (`eval/results/accessibility.json`).
- **Agent answers** cite IDs so an engineer can check every claim against the map.

9. Responsible use

Risk	What we do
Privacy of people in dashcam frames	Faces (YuNet) and plates/text are blurred **before** any frame or evidence image is stored; unblurred frames never leave the container. Survey data are synthetic; real images are used only from licensed research datasets
Automation without accountability	The agent cannot file anything. Every work order needs a named human approval (Step Functions task token, stored server-side); rejections are recorded and counted (override rate on the dashboard)
Hallucinated or fabricated defects	Three layers: tools reject unknown IDs, a database trigger rejects packages citing unknown defects, the final answer is checked for unknown IDs (hard gate in CI)

Risk	What we do
Over-claiming	Depth is never estimated (materials use a stated thickness); "repaired" means "not seen again" and is labelled as such; the real-photo failure is reported prominently; the map says the data are simulated
Fairness of prioritisation	The exposure score weights schools, hospitals and bus stops, which could under-serve areas with fewer mapped facilities in OpenStreetMap. It is deterministic and visible in every work order, so it can be audited and changed; engineers can override and overrides are tracked
Unintended filing with a real authority	Filing goes only to a mock endpoint (`MOCK-⋯` tickets)
Data and model licences	Pothole Mix is research-only, so the SegFormer weights are not published; a licence-clean model trained only on our synthetic data is released (DATA-LICENSES.md)
Cost and abuse of the cloud	Credits-only operation with a real-spend alarm; Cognito with no self sign-up; API throttling; only first-party Amazon models

10. Limitations and next steps

1. **Real-world detection**: label pixel masks on real Indian dashcam frames (the gap in §7.4), then re-measure on real drives with tape-measured potholes.
2. **Verification**: fewer false detections (the main source of wrong links), and a patch-aware repair check.
3. **Scale**: more drives per road (8 were planned, 2 evaluated) to measure degradation rates.
4. **Not used**: AgentCore Gateway and Memory (tools run inside the agent container).

11. Reproduce

See BUILD-DEPLOY-TEST.md: local quick start (no AWS), full AWS deployment in ten steps, and every test and evaluation command. Design decisions with reasons: `main_docs/plan/11-decisions-and-risks.md` (D-01 to D-40).